



1. SYSTEM HEALTH & MODULE MATRIX

Module / Subsystem	Status	Diagnostic Findings & Impact Summary
Build & Compilation Vite 7.3 / Rollup	FIXED	Unclosed JSX <div> in SatDashboard.jsx resolved. npm run build builds cleanly in 14.1s with PWA assets generated.
Cloud Database Supabase Cloud API	BROKEN	.env.local contains Vercel-prefixed variables (hffdashboardfinal_) while code requires VITE_. DNS for trirtnxcyjwdzhjmaubn.supabase.co returns ENOTFOUND (paused in console).
Local Storage (Dexie) IndexedDB (v1-v12)	WORKING	Robust offline-first architecture. Sequential schema migrations v1-v12, reactive UI hooks (useLiveQuery), and local CRUD transactions operate cleanly without errors.
Campaign Hub PersonList & Attendance	WORKING	Directory search & filtering by type/affiliation, interactive 12-day attendance matrix, books distributed tracking, and facilitator linkage function properly.
Facilitator Portal Dashboard & Onboarding	WORKING	Role-based gateway, 8-question demographic onboarding, participant linkage to facilitator UUID, and attendance tracking work seamlessly.
Data Deduplication MaintenanceTool	WORKING	Fuzzy detection based on Name + Age + Affiliation. Dry-run previews and non-destructive merging into master records operate reliably.
SAT Certificates pdf-lib + JSZip	BLOCKED	PDF stamp logic and certificate template (public/certificate_blank.pdf) are ready, but blocked by the JSX compilation error in SatDashboard.jsx.
Overview Dashboard Metrics & Charts	PARTIAL	Core KPI cards and charts render, but the Gender Filter buttons are not wired to data, and "Download Summary" has no click handler.
Data Ingestion Excel/CSV Parser	ORPHANED	Robust parser (hffRegister.js) and FileUpload.jsx exist, but have no button or interface mounted in the active dashboard UI.

2. WHAT IS WORKING WELL

1. Offline-First Resilience (Dexie.js)

12-version IndexedDB schema provides reliable offline caching for registrations, participants, and notice bulletins with live reactivity across all views via useLiveQuery.

2. Comprehensive Campaign Hub

Supervisors can filter hundreds of records, inspect facilitator cohorts, record attendance across all 12 campaign days, and track educational book disbursements.

3. Automated Data Deduplication

The maintenance engine identifies duplicate participants across identical names, ages, and affiliations, combining contact details and attendance records into the most recent master entry.

4. Dedicated Facilitator Portal

Field facilitators logging in without completed onboarding are automatically guided through demographic profiles before accessing their mobile attendance sheet.

5. Two-Way Sync Architecture

Differential synchronizer compares timestamps (updated_at) to prevent pulling unchanged records, propagates soft deletes (is_deleted), and queues local offline changes.

6. Client-Side Certificate Engine

Built-in pdf-lib integration stamps formatted participant names directly onto the high-resolution certificate template and bundles qualifying attendees into ZIP files.

3. CRITICAL ISSUES & BROKEN COMPONENTS

Issue A: Syntax Error Halts Build in SatDashboard.jsx

CRITICAL BLOCKER

In `src/components/SatDashboard.jsx`, line 225 opens a container `div` for the search input and export buttons, but never closes it before line 257. This leaves the outer card `div` unclosed, terminating the build process with error code 1.

Line 225: `<div className="flex flex-col sm:flex-row justify-between gap-4">` Line 226-237: `<div className="relative flex-1 max-w-md">...</div>` Line 238-255: `<div className="flex justify-end gap-3">...</div>` + ADD HERE: `</div>` **Missing closing tag for Line 225** → Line 257: `<div className="glass-card p-8 bg-white/50">`

Issue B: Supabase Environment Variable Mismatch & Inactive Host

AUTH & CLOUD DOWN

- Key Name Mismatch:** `.env.local` contains Vercel-generated variables (`NEXT_PUBLIC_hffdashboardfinal_SUPABASE_URL`, `hffdashboardfinal_SUPABASE_ANON_KEY`), while `src/lib/env.js` expects `VITE_SUPABASE_URL` and `VITE_SUPABASE_ANON_KEY`. As a result, `isConfigured` evaluates to `false`.
- DNS Unresolved:** Network lookup for `trirtnxcyjwdzhjmaubn.supabase.co` returns `ENOTFOUND`, indicating that the free-tier Supabase project has been paused due to inactivity.

Resolve-DnsName : `trirtnxcyjwdzhjmaubn.supabase.co` : DNS name does not exist **Fix:** Log into `supabase.com/dashboard` and click "Restore Project" / "Unpause".

Issue C: Disconnected UI Controls in Dashboard.jsx

USER EXPERIENCE BUG

- Gender Filter Placebo:** The Universal / Male / Female toggle buttons update a local state variable (`genderFilter`), but this value is never passed to `AttendanceChart` or used to filter `dailyStats`.
- Dead "Download Summary" Action:** In `Dashboard.jsx`, `<ActionPanel />` is instantiated without passing an `onDownload` callback, leaving the prominent button completely non-functional.
- Static Dummy Notice Board:** `Dashboard.jsx` imports an outdated mock (`NoticeBoard.jsx` with fake notices like "Maun Site Visit Oct 14") instead of the live dynamic hub/`NoticeBoard.jsx`.

Issue D: Serverless API Schema Discrepancies

BACKEND WARNING

- api/health.js:** Attempts to query a non-existent `notes` table, returning a `503 Service Unavailable` error on health checks.
- api/process-registration-queue.js:** Queries `processed = false` on the `registrations` table, but that column does not exist in the Supabase schema.
- scripts/calculate_facilitator_certificates.js:** Uses a relative path pointing to `scripts/.env.local` instead of `../.env.local`.

4. PRIORITIZED REMEDIATION PLAN

PRIORITY	TASK	IMPLEMENTATION DETAILS & FILES IMPACTED
RESOLVED	Fix SatDashboard.jsx Syntax	[FIXED] Closed missing <code></div></code> on line 256. Verified full production build succeeds cleanly.
P1 - HIGH	Restore Supabase Connection	Add standard <code>VITE_SUPABASE_URL</code> aliases to <code>.env.local</code> and <code>env.js</code> . Unpause project in Supabase cloud dashboard.
P2 - MED	Connect Dashboard Controls	Pass <code>genderFilter</code> into attendance calculation. Bind <code>onDownload</code> to trigger deep analysis PDF export in <code>Dashboard.jsx</code> .
P2 - MED	Unify Notice Board	Replace static dummy <code>src/components/NoticeBoard.jsx</code> with dynamic <code>src/components/hub/NoticeBoard.jsx</code> on the overview dashboard.
P2 - MED	Mount Excel File Upload	Add a button in Campaign Hub to open <code>FileUpload.jsx</code> and allow administrators to ingest Excel campaign registers.
P3 - LOW	Align Serverless Endpoints	Update <code>api/health.js</code> to probe <code>registrations</code> instead of <code>notes</code> . Fix script path in <code>calculate_facilitator_certificates.js</code> .
P4 - CLEAN	Repository Pruning	Delete legacy Next.js files (<code>src/app/</code> , <code>next.config.mjs</code> , <code>.next/</code>) and duplicate <code>src/lib/server/</code> directory.